



FINAL REPORT

MATH336

MATHEMATICAL MODELLING

THE SEARCH FOR NEW WORLDS

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1 Abstract

This project investigates exoplanet detection using light curve data from the Kepler mission, combining astrophysical modelling with machine learning techniques. Using the transit method, we identify potential exoplanets based on periodic dips in stellar brightness. Raw pixel data undergo calibration, aperture photometry, and noise correction, facilitated by the Lightkurve library.

We implement two detection approaches. The first is a distance-based classifier using Dynamic Time Warping (DTW), which aligns and compares light curves while accounting for timing distortions. To reduce computational cost, a filtering step based on stellar parameters narrows down the search space. The second approach uses a 2D Convolutional Neural Network (CNN), applied to image-like representations of phase-folded light curves.

Our final CNN model achieved 99.2% accuracy and an AUC score of 0.97. Data preprocessing steps such as standardisation, class balancing, and augmentation improved the model’s performance. While the model was generally effective, some misclassifications occurred due to binary star signals or instrumental noise.

The project shows how machine learning can aid in handling large astronomical datasets and supports the development of scalable tools for exoplanet detection. Future improvements could focus on better generalisation across missions and addressing class imbalance.

Literature Review

The detection of exoplanets via the transit method — where periodic dips in stellar brightness indicate planetary transits — has transformed astronomy. Since the Kepler mission launched in 2009, exoplanet detection has rapidly expanded, with K2, TESS, and the forthcoming PLATO mission increasing the data volume exponentially [Singh & Misra, 2020]. With millions of light curves now available, manual analysis has become infeasible, prompting the need for automated detection systems [Malik et al., 2020].

Classical Machine Learning Approaches

Initial detection pipelines used classical machine learning, relying heavily on engineered features. McCauliff et al. developed Autovetter, a random forest model using features such as transit depth, duration, and various statistics [McCauliff et al., 2015]. Though effective, its performance was constrained by hand-crafted features. Robovetter, introduced by Thompson et al., applied decision trees with domain rules to exclude binary star signals [Thompson et al., 2018]. These tools improved over manual vetting but lacked flexibility for novel or noisy data [Mena et al., 2019].

Deep Learning Revolution

Deep learning’s rise led to models capable of learning features directly from raw light curves. Shallue and Vanderburg introduced AstroNet, a dual-branch convolutional neural network (CNN) processing both global and local views of phase-folded data [Shallue & Vanderburg, 2018]. AstroNet outperformed classical models by extracting complex temporal patterns. Building on this, Burhanudin et al. used LSTMs and stellar parameters to further enhance prediction quality [Burhanudin et al., 2021].

Salinas et al. proposed a transformer-CNN hybrid integrating attention mechanisms and centroid data, improving artefact discrimination and interpretability [Salinas et al., 2022]. Similarly, Priyadarshini and Puri demonstrated a stacked CNN ensemble achieving 99.62% accuracy on Kepler data [Priyadarshini & Puri, 2021].

Data Representation and Preprocessing

Tiensuu et al. showed that encoding light curves as image-like matrices — using Gramian Angular Fields (GAF), Markov Transition Fields (MTF), and Recurrence Plots — improved CNN performance compared to SVMs (recall: 1.000 vs. 0.800) [Tiensuu et al., 2019]. Their results highlighted the critical role of preprocessing, including detrending, smoothing, and Fourier transforms.

Jin et al. explored unsupervised methods like k-means clustering, demonstrating that effective classification is possible even without labels, reporting accuracies between 88.5% and 99.79% across various algorithms [Jin et al., 2022].

Cross-Mission Applications

A key challenge is generalisation across missions. Malik et al. built a pipeline achieving 98% accuracy on TESS and an AUC of 0.948 on Kepler data, despite the noisier, shorter-duration TESS light curves [Malik et al., 2020].

Beyond detection, models now target higher-order tasks. Singh and Misra used ML to estimate planetary habitability from orbital and stellar parameters [Singh & Misra, 2020]. Mena et al. automated classification of transit morphologies, enhancing system categorisation [Mena et al., 2019].

Challenges and Future Work

The major hurdle remains class imbalance: true planets are vastly outnumbered by false positives [Tiensuu et al., 2019]. Solutions include oversampling, anomaly detection, and prioritising recall. Interpretability is also critical, as astronomers seek physically grounded insights. Recent work has added physics-informed constraints to model architectures [Salinas et al., 2022].

Looking ahead, with missions like PLATO and JWST, detection pipelines must scale efficiently while remaining sensitive to rare, Earth-like planets. Semi-supervised learning, cross-mission transfer learning, and real-time systems will define the next era in exoplanet detection.

2 Introduction

2.1 Exoplanets

Are we alone in this world? This one question has haunted scientists and enthusiasts alike for many years. Exoplanets (or extrasolar planets) are planets that exist outside our Solar System, orbiting stars other than the Sun. Thousands of exoplanets have been discovered, and they vary widely in size, chemical composition, and orbital characteristics. These planets are typically found in stellar systems, which simply means that they orbit a star. This is because planets are a by-product of stellar formation, created from the protoplanetary gas disk surrounding the star in its early stages.

2.2 Origins

Apart from the elegance and mathematical beauty involved in the study of celestial mechanics, research in this field yields profound insights of the origins of life and possibly alien life as well. Some exoplanets are discovered in the 'habitable zone' of their stellar systems, which means that they are at the right distance from the star (this distance depends on the type of star in question) for liquid water to be present, which we consider to be a necessary condition for life. Moreover, spectroscopy-based methods allow us to infer the chemical compositions of these planets, deepening our understanding of adjacent fields like atmospheric science and climate as well. However, before research in all the various subfields pertaining to exoplanets can commence, we first need methods for their discovery.

2.3 Methods for Detection

The detection of exoplanets involves various observational techniques, each with its strengths and limitations. Below, we summarise the primary methods used for exoplanet discovery:

1. **Transit Method**

Detects dips in starlight as a planet passes in front; reveals size and atmosphere but requires precise alignment.

2. **Radial Velocity (Doppler) Method** - Measures star's spectral shifts due to orbiting planets; effective for mass estimates, but biased toward large planets.

3. **Direct Imaging** - Captures actual images of exoplanets using high-contrast techniques; works best for large, distant planets.

4. **Gravitational Microlensing** - Uses light bending by a planet-hosting star to detect distant or rogue planets; events are rare and not repeatable.
5. **Astrometry** - Tracks tiny shifts in a star's position due to planetary pull; precise but technologically demanding.
6. **Pulsar Timing** - Detects timing variations in pulsar radio pulses; extremely accurate but limited to rare pulsar systems.
7. **Orbital Phase Variations** - Observes brightness changes due to reflected light from planets; effective for close, reflective planets.

Among the methods discussed above, the transit method has been most successful, not only in terms of detection volume but also the potential for post-detection inference. Let us examine this method in more detail in the next section.

2.4 The Transit Method

When an exoplanet crosses our line of sight to its host star, the apparent brightness of the star is reduced. We call this phenomenon a transit event, and if such a dip in brightness of the star is observed periodically, we can deduce the presence of an exoplanet in its orbit. Figure 5 illustrates this method. As one might imagine, the length of the transit or dip in brightness directly corresponds to the orbital period of the planet, while the depth of the dip corresponds to its radius.

The Kepler mission by NASA was aimed at using this method to discover new exoplanets, and the data from this mission was made publicly available via the Mikulski Archive for Space Telescopes (MAST) database. As of now, our project is focused exclusively on the data collected from the Kepler mission, but we might generalise the models' capabilities to other missions like K2 and TESS in the future. We have also augmented our dataset using data from the NASA Exoplanet Archive.

3 Data Engineering

From Raw Target Pixel Data to Flux Extraction

3.1 Raw target Pixel Data

The raw data from telescopes such as TESS and Kepler consists of target pixel files (TPFs). These telescopes capture light from a distant star and store that information in TPFs. They contain the following:

- **Pixel Intensity Values:** Each pixel represents a small region of the sky, with its intensity indicating the amount of light received.

- **Time Stamps:** Each data point is tagged with the exact observation time.
- **Background Noise:** Unwanted signals from cosmic rays, spacecraft jitter, or detector noise are also present.

They are 2D arrays of pixel values over time. Let: $I(x, y, t)$ be the intensity (in electrons per second) at pixel coordinates (x, y) and time t .

$N_x \times N_y$ be the dimensions of the pixel array.

N_t be the number of time steps.

The raw data is affected by:

- **Photon noise:** Poisson noise due to the discrete nature of photons.
- **Read noise:** Gaussian noise introduced during the readout process.
- **Systematic effects:** Scattered light, spacecraft jitter, and thermal variations.

3.2 Pixel Calibration

Before extracting flux, the raw pixel values must be calibrated to remove instrumental effects.

Background Subtraction

The background $B(x, y, t)$ is estimated from pixels not containing the target star. The calibrated intensity $I_{\text{cal}}(x, y, t)$ is:

$$I_{\text{cal}}(x, y, t) = I(x, y, t) - B(x, y, t)$$

Example: If $I(x, y, t) = 1000$ electrons/s and $B(x, y, t) = 200$ electrons/s, then:

$$I_{\text{cal}}(x, y, t) = 1000 - 200 = 800 \text{ electrons/s}$$

Flat-Field Correction

In real imaging systems (like CCDs or CMOS sensors), not all pixels respond identically to incoming light — some pixels may be slightly more or less sensitive due to manufacturing imperfections, dust particles, or optical distortions. The flat field corrects for these variations. It is a multiplicative factor. Each pixel has a different sensitivity or flat field $F(x, y)$. The corrected intensity $I_{\text{flat}}(x, y, t)$ is:

$$I_{\text{flat}}(x, y, t) = \frac{I_{\text{cal}}(x, y, t)}{F(x, y)}$$

3.3 Aperture Photometry

The flux from the target star is extracted by summing the intensities within an aperture (a region of pixels containing the star). It is the same aperture as that is talked about in photography or camera technology. Let:

A be the set of pixels in the aperture.

The raw flux $F_{\text{raw}}(t)$ at time t is:

$$F_{\text{raw}}(t) = \sum_{(x,y) \in A} I_{\text{flat}}(x, y, t)$$

Aperture Optimization

The chosen aperture must:

- Capture most of the target star's flux.
- Minimise background noise and contamination from nearby stars.

The optimal aperture A^* is found by maximizing the following ratio:

$$A^* = \arg \max_A \left(\frac{\sum_{(x,y) \in A} I_{\text{flat}}(x, y, t)}{\sum_{(x,y) \in A} \sigma^2(x, y, t)} \right)$$

Where:

- $I_{\text{flat}}(x, y, t)$ = Calibrated intensity at pixel (x, y) at time t .
- $\sigma(x, y, t)$ = Noise/uncertainty for that pixel (e.g., read noise, photon noise).

Example Calculation

Suppose the following 5 pixels are candidate aperture pixels:

(x, y)	$I_{\text{flat}}(x, y, t)$	$\sigma(x, y, t)$
(1, 1)	500	10
(1, 2)	450	12
(2, 1)	470	11
(2, 2)	430	13
(3, 2)	460	9

Finally, the optimised ratio is:

$$\text{SNR Ratio} = \frac{2310}{615} \approx 3.76$$

In this example, the optimal aperture achieves a signal-to-noise ratio of approximately 3.76.

To be considered:

- Small Aperture: Reduces contamination but may lose some flux.
- Large Aperture: Captures more flux but risks adding noise from background sources.

3.4 Noise Modelling and Detrending

The raw flux $F_{\text{raw}}(t)$ contains systematic noise $N_{\text{sys}}(t)$ (e.g., scattered light, spacecraft jitter) and random noise $N_{\text{rand}}(t)$. The observed flux $F_{\text{obs}}(t)$ is:

$$F_{\text{obs}}(t) = F_{\text{true}}(t) + N_{\text{sys}}(t) + N_{\text{rand}}(t)$$

Systematic Noise Model

Systematic noise is modeled using a design matrix $X(t)$ and coefficients β :

$$N_{\text{sys}}(t) = X(t) \cdot \beta$$

Principal Component Analysis (PCA)

To reduce dimensionality and remove stochastic noise, PCA is applied to the design matrix:

$$X = U \cdot \Sigma \cdot V^T$$

Example: If X has 100 regressors, PCA might reduce it to 5 principal components that capture 95% of the variance.

Corrected Flux

The corrected flux $F_{\text{corr}}(t)$ is obtained by subtracting the modeled systematic noise:

$$F_{\text{corr}}(t) = F_{\text{obs}}(t) - X(t) \cdot \beta$$

Uncertainty Quantification

The uncertainty in the corrected flux $F_{\text{corr}}(t)$ is propagated from the uncertainties in the observed flux and the regressors:

$$\sigma_F^2 = \sigma_{\text{obs}}^2 + \sum_{i=1}^k \left(\frac{\partial F_{\text{corr}}}{\partial X_i} \cdot \sigma_{X_i} \right)^2$$

Residual Analysis

The residuals $\epsilon(t)$ are the difference between the corrected flux and the true flux:

$$\epsilon(t) = F_{\text{corr}}(t) - F_{\text{true}}(t)$$

3.5 Instrumental Effects and the Role of the Pixel Response Function

After detrending, we assume that the flux values used in our model reflect the intrinsic stellar brightness. However, from a physical standpoint, the observed flux $F(t)$ is the outcome of a more complex photon detection process, which can be mathematically expressed as:

$$F(t) = \iint I(x, y, t) \cdot \text{PRF}(x, y) dx dy \quad (1)$$

Here, $I(x, y, t)$ denotes the time-dependent intensity distribution of the star on the detector, and $\text{PRF}(x, y)$ is the Pixel Response Function. The PRF describes how sensitive each point on the detector is to incident light, incorporating the optical and electronic characteristics of the instrument.

The PRF is inherently linked to the Point Spread Function (PSF), which characterises how light from a point source spreads on the detector due to diffraction and optical aberrations. This relationship is often expressed as a convolution of the PSF with pixel-level response characteristics, indicating that the flux we observe is shaped by both astrophysical signals and instrumental properties.

As a result, the flux used to train our machine learning model is not solely determined by the astrophysical transit event but also by how the telescope records and distributes this light. While the Kepler pipeline employs polynomial fitting techniques to approximate and correct for these instrumental effects, residual systematics remain. These systematics are implicitly learned by the classifier during training

The `Lightkurve` library automates much of the process described above.

4 Distance Based Classifier

A distance-based classifier is a simple yet effective algorithm for determining whether a star is an exoplanet host or a non-host based on its light curve data. The classification process follows these steps:

- The dataset is partitioned into training and testing sets, with the Kepler Object of Interest (KOI) disposition serving as the target variable. This variable has values 0 and 1, indicating non-host and host star.

- For each test example light curve, the algorithm computes its ‘distance’ from all training examples and identifies the ‘nearest’ one.
- The test example is then assigned the same class as its nearest neighbour in the training set.

This method relies on the assumption that similar light curves correspond to similar astrophysical characteristics, making it a natural choice for exoplanet classification.

4.1 Measuring Distance Between Two Light Curves

When comparing light curves to classify stars as potential exoplanet hosts, a key challenge is determining how ‘similar’ two light curves are. A naive approach is to measure the Euclidean distance between corresponding points in the two time series. However, this method assumes that the two curves are perfectly aligned in time, which is often not the case.

In reality, astronomical observations are subject to variations in observation conditions, instrumental noise, and intrinsic stellar activity. These factors can cause slight shifts in time or distortions in the recorded light curve. As a result, two light curves representing the same underlying phenomenon may appear misaligned, leading to misleading distance measurements.

To address this, we need a more flexible approach—one that accounts for variations in timing by allowing for slight stretching or compression of the time axis. This ensures that we compare meaningful features rather than being misled by superficial misalignments.

Figure 6 illustrates an alternative distance metric to the standard Euclidian distance, which computes the distance between two series after an ‘optimal’ alignment. Before we discuss how this time-warped distance metric is defined, we need to talk about subordinated processes.

4.2 Dynamic Time Warping (DTW)

Dynamic Time Warping (DTW) can be viewed as a discrete subordinated process, where one sequence is “warped” in time to optimally align with another. In probability theory, a subordinated process is one in which a primary process is indexed by a second process, which itself is a non-decreasing sequence of random time steps. Similarly, in DTW, the warping function plays the role of the subordinator, determining the mapping between the time indices of two sequences.

Unlike standard Euclidean distance, which assumes a one-to-one correspondence between elements in time series, DTW allows flexible alignments by stretching or compressing segments as needed. This makes it particularly useful for time-dependent data where sequences may exhibit similar patterns but at different paces.

4.2.1 Given Time Series

We consider two discrete time series:

$$X = (x_1, x_2, \dots, x_N), \quad Y = (y_1, y_2, \dots, y_M) \quad (2)$$

Each element x_i and y_j represents an observation at time index i and j , respectively. The goal of DTW is to find an optimal alignment between these sequences.

4.2.2 Cost Matrix

The local cost matrix measures the dissimilarity between elements of X and Y . A common choice is the squared Euclidean distance:

$$D(i, j) = (x_i - y_j)^2 \quad (3)$$

This matrix D quantifies how different each pair of points is.

4.2.3 Cumulative Cost Matrix

The cumulative cost matrix $C(i, j)$ defines the optimal alignment cost up to position (i, j) . It is recursively computed as:

$$C(i, j) = D(i, j) + \min(C(i-1, j), C(i, j-1), C(i-1, j-1)) \quad (4)$$

where:

- $C(i-1, j)$ represents an insertion (skipping an element in X),
- $C(i, j-1)$ represents a deletion (skipping an element in Y),
- $C(i-1, j-1)$ represents a match (aligning x_i with y_j).

This recurrence ensures that the optimal path minimises the total cost over all possible alignments.

4.2.4 DTW Distance

The final DTW distance is given by:

$$DTW(X, Y) = \sqrt{C(N, M)} \quad (5)$$

where $C(N, M)$ is the minimum cumulative cost at the end of the alignment path. The square root is optional and depends on whether one wants to maintain units comparable to Euclidean distance.

4.3 Integrating DTW in the Classification Pipeline

In practice, for two lightcurves of lengths m and n , DTW has a time complexity of $O(m \cdot n)$. In the broader context of our classification pipeline, if our training set has p examples and testing set has q examples, then the time-complexity over the composite algorithm becomes $O(m \cdot n \cdot p \cdot q)$, which is simply computationally infeasible for large datasets and long time-series.

To ameliorate this problem, we apply a hierarchical pre-search conditional filter. This involves extracting statistical features from the time-series, and filtering for the 'k' nearest neighbours in this feature space, and then applying DTW to each of these neighbours that passed the filter.

4.3.1 Performance

We'll start by analyzing the performance of the baseline model, i.e. KNN with no post-filter DTW search. Using $K=10$, the Manhattan Distance metric, and distance-weighted voting, we obtained the following results:

Table 1: Classification Report

Class	Precision	Recall	F1-score	Support
0	0.96	0.97	0.97	1204
1	0.75	0.65	0.70	148
Accuracy	0.94 (on 1352 samples)			
Macro avg	0.85	0.81	0.83	1352
Weighted avg	0.93	0.94	0.94	1352

As you can see, due to the heavy class imbalance, the model struggles balance precision and recall, specifically for the positive (1) class. Our hope is that by using the post-filter DTW search, we'll be able to make the process of delineating light curves surgical, and hence increase our recall, while maintaining or possibly even lowering the false positives, thereby increasing precision.

Now lets analyze the performance of the DTW integrated classifier.

Table 2: Classification Report for DTW-based Hierarchical KNN (Refined Presentation)

Class	Precision	Recall	F1-score	Support
Class 0 (Non-exoplanet)	0.97	0.98	0.98	1204
Class 1 (Exoplanet)	0.82	0.75	0.78	148
Overall Accuracy	0.95 (on 1352 samples)			
Macro Average	0.89	0.86	0.88	1352
Weighted Average	0.95	0.95	0.95	1352

As you can see, the use of DTW in the post-search stage resulted in significant improvements in classification metrics, without sacrificing precision.

5 Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are a class of deep learning models primarily used for analysing structured data such as images and time-series signals. They automatically learn to detect patterns and features in the data, making them effective for a range of classification and detection tasks.

6 Light Curve Analysis Using CNNs

To utilise CNNs, these time-series data were transformed into image-like representations through techniques called phase folding. These visualisations allow CNNs to detect meaningful features that may indicate the presence of exoplanets or other stellar phenomena. This removes the need for manual feature extraction and enables the network to autonomously learn patterns associated with planetary transits.

6.1 Initial Exploration with 1D CNN

We initially explored a 1D CNN architecture that processed raw light curve time-series directly. While this provided early insights, we transitioned to a 2D CNN model using image representations of the data to capture more complex spatial patterns and improve classification performance.

7 Mathematical Foundations of CNNs

7.1 Convolutional Layers

Convolutional layers apply filters over the input to produce feature maps that capture local patterns. For a 2D input I and a filter K :

$$S(i, j) = (I * K)(i, j) = \sum_m \sum_n I(i + m, j + n) \cdot K(m, n)$$

7.2 Activation Functions

ReLU (Rectified Linear Unit) introduces non-linearity and is defined as:

$$\text{ReLU}(x) = \max(0, x)$$

7.3 Pooling Layers and Fully Connected Layers

Max pooling reduces dimensionality and extracts dominant features:

$$P(i, j) = \max_{(m, n) \in \text{window}} S(i + m, j + n)$$

These layers integrate the extracted features to perform final classification.

8 Model Architecture (2D CNN)

- Convolutional layers with increasing filters (32, 64, 128), each followed by ReLU and max pooling.
- Batch normalisation and dropout to stabilise training and prevent overfitting.
- Flatten and fully connected dense layers.
- Sigmoid activation in the final layer for binary classification.

9 Model Performance, Evaluation, and Data Handling

Data Preprocessing and Augmentation

- All input data was standardized (zero mean, unit variance).
- Positive (exoplanet-hosting) cases were oversampled to address class imbalance.
- Random temporal shifts were applied to simulate data variability and improve generalization.

Performance and Evaluation

Our model achieved an impressive **99.2% accuracy**, highlighting its effectiveness in distinguishing exoplanet-hosting stars.

Loss and Accuracy

Training and validation loss steadily decreased, while accuracy consistently improved, indicating effective learning and generalisation.

Optimal Cutoff Score

- The chosen threshold of 0.67 provides a strong balance between sensitivity and specificity.
- AUC score of 0.97 demonstrates robust classification performance.

Misclassified Stars

Despite high accuracy, some stars were incorrectly classified as hosting exoplanets due to similar patterns caused by other factors:

- Binary eclipses (eclipsing binary stars).
- Instrumental noise or anomalies.

10 Conclusion

CNNs offer a powerful, automated approach for classifying stellar phenomena using light curve data. By converting time-series into image formats, we leveraged a 2D CNN that achieved 99.2% classification accuracy. While limitations such as class imbalance and astrophysical ambiguity remain, this method holds strong promise for scalable exoplanet detection in large datasets.

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Appendix: Figures

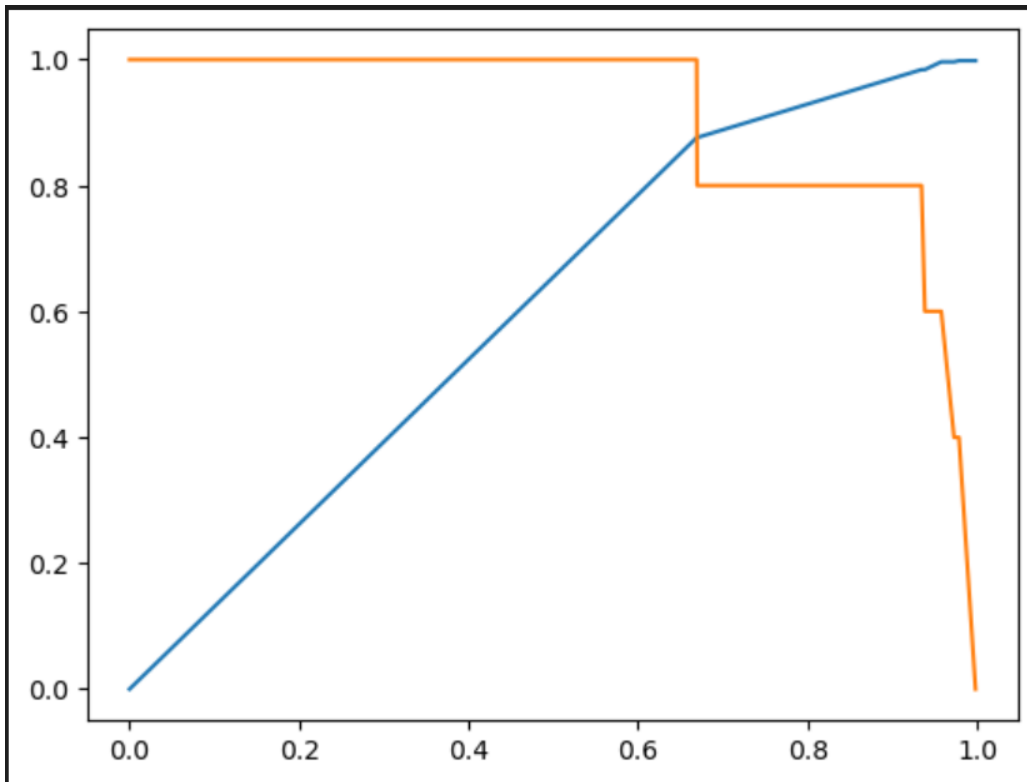


Figure 1: Optimal Cutoff Score

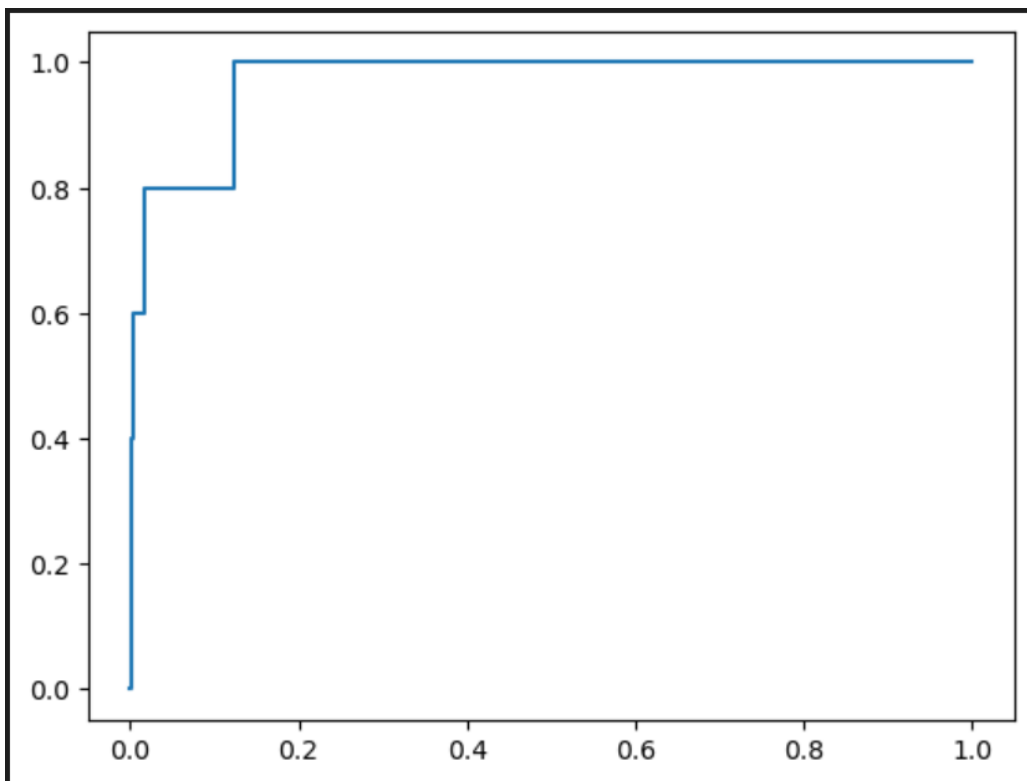


Figure 2: Optimal Cutoff Score

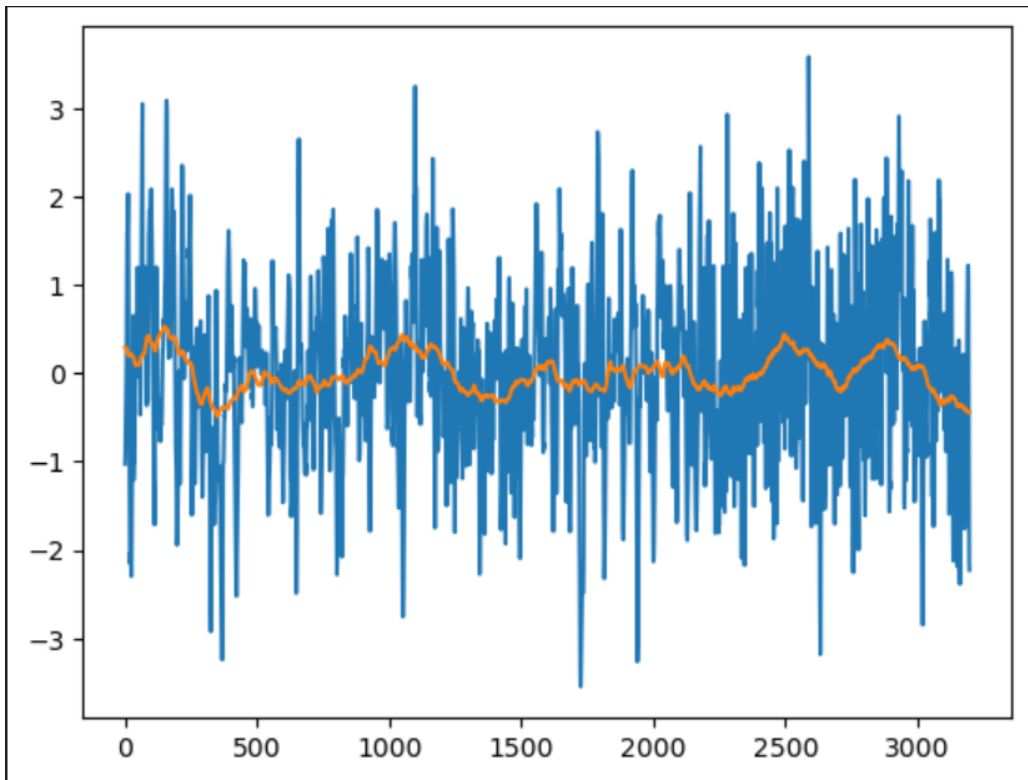


Figure 3: Misclassified Star 1

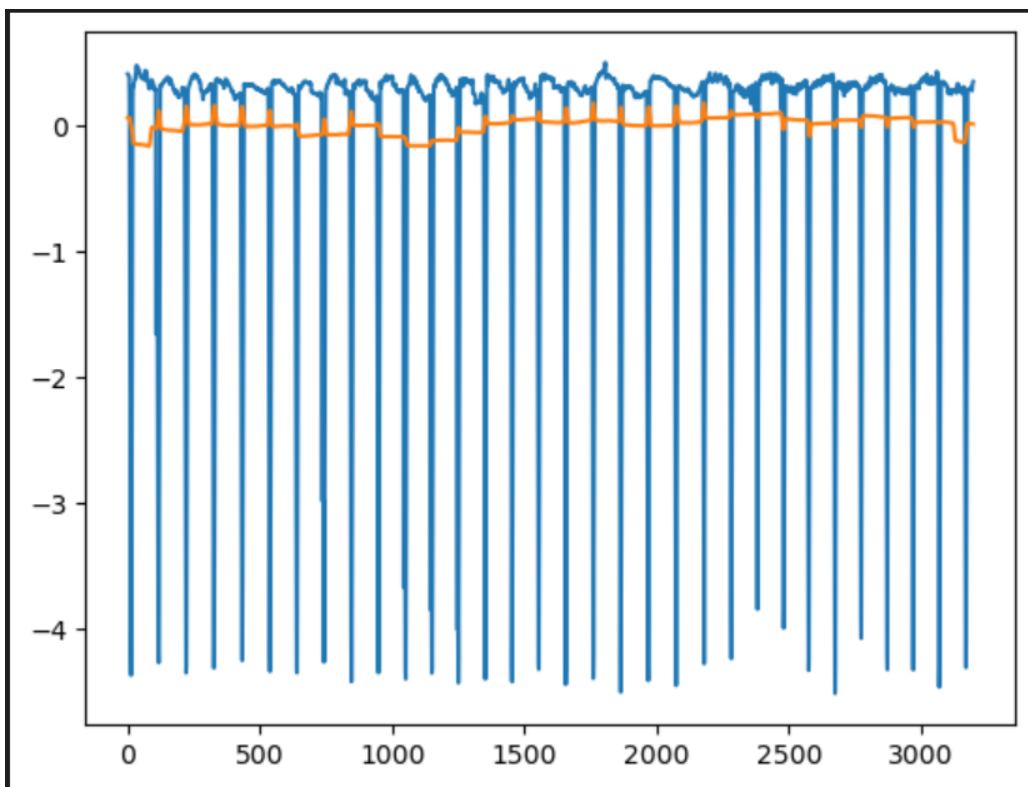


Figure 4: Misclassified Star 2

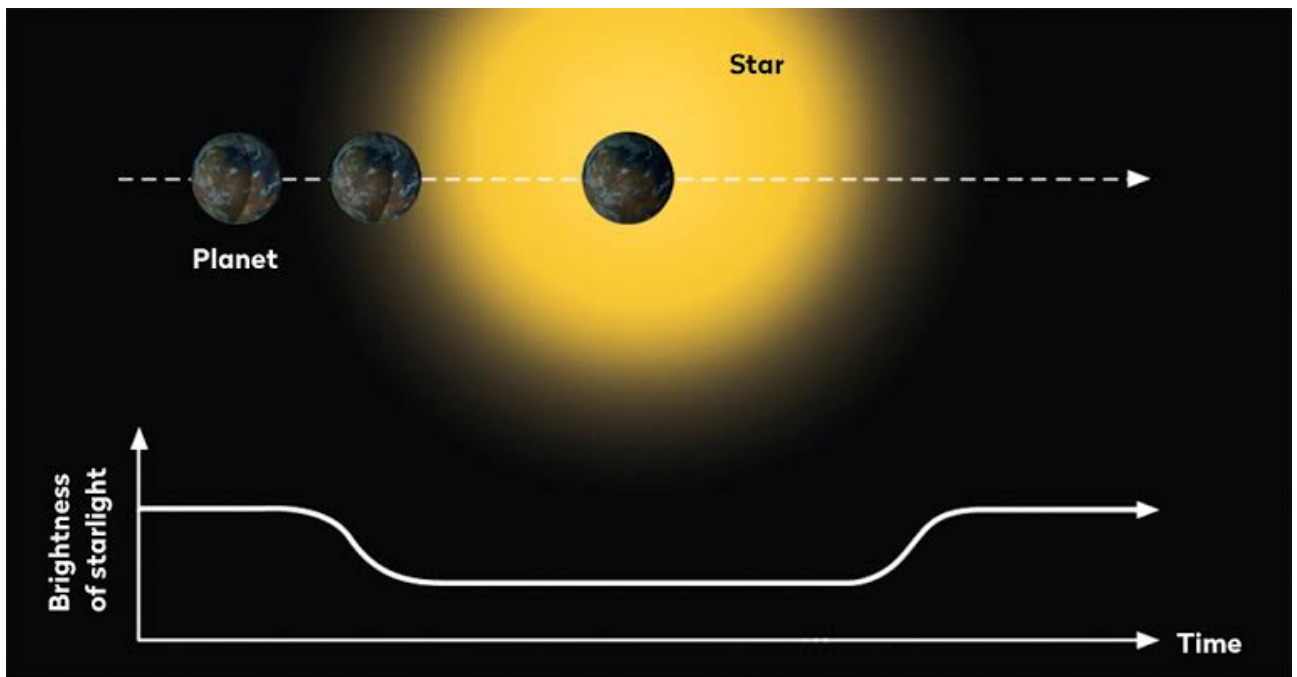


Figure 5: Lightcurve of a Transit Event

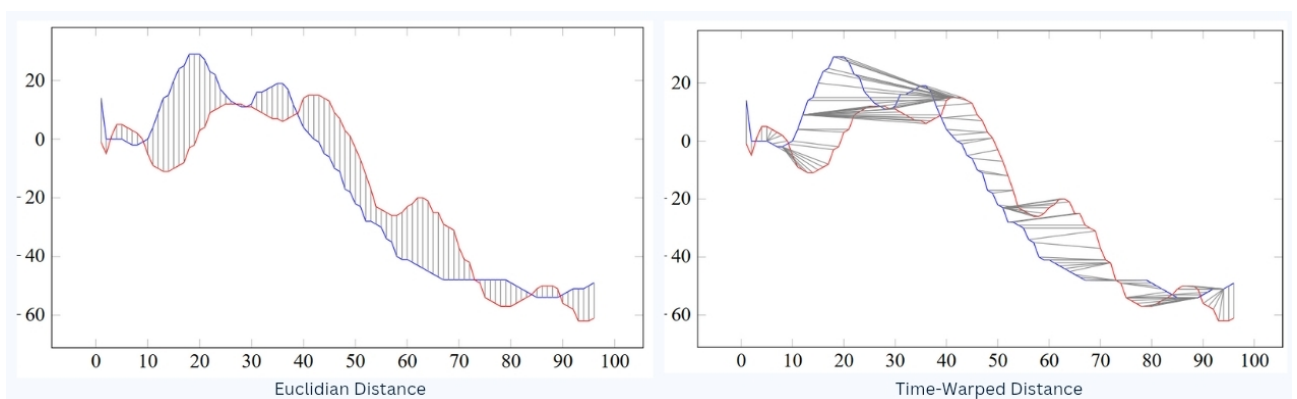


Figure 6: Dynamic Time Warping

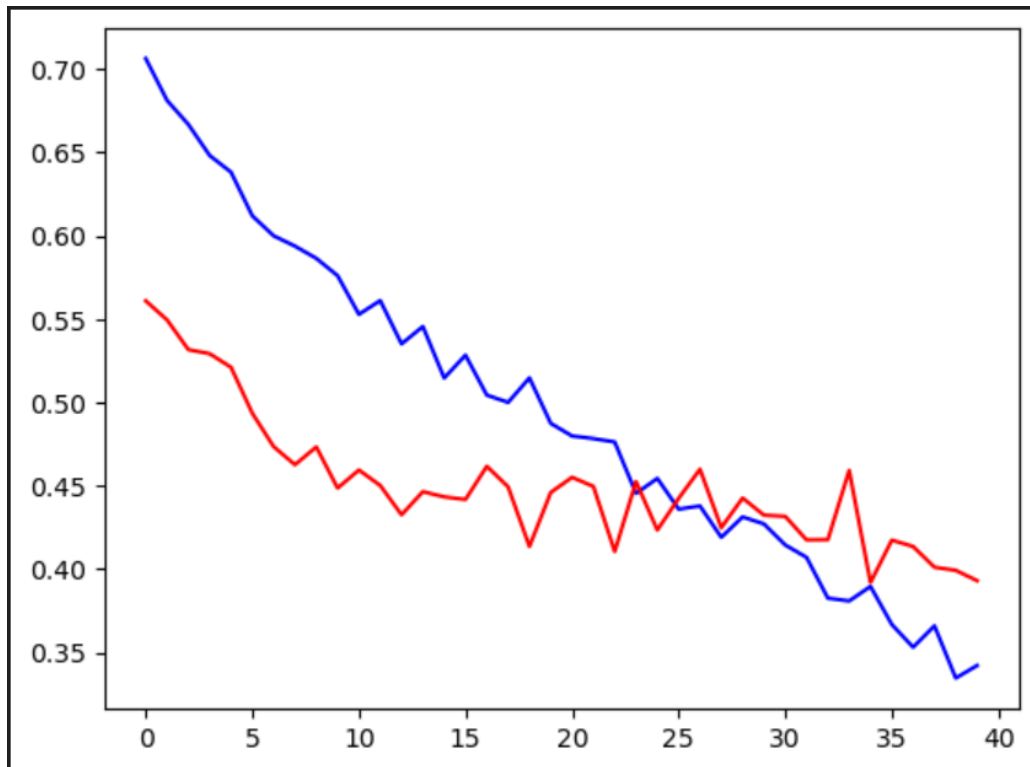


Figure 7: Training and Validation Loss

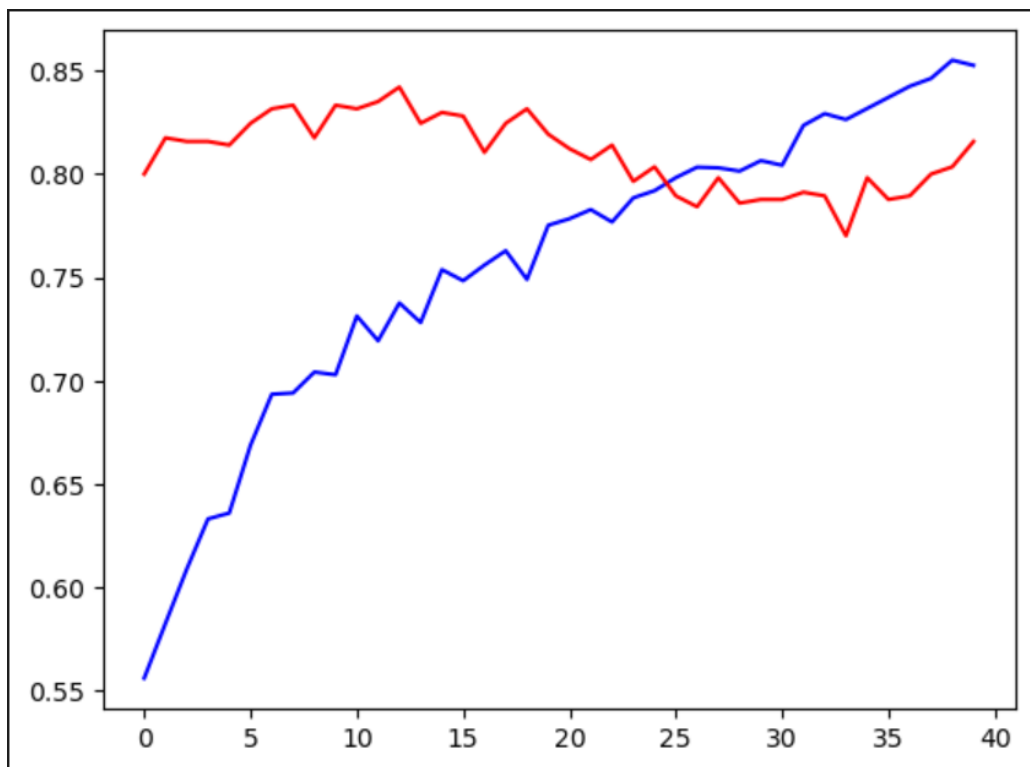


Figure 8: Training and Validation Accuracy